

Kaushik Gunasekaren

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PERSONAL SUMMARY

Driven undergraduate at Arizona State University pursuing a Bachelor's degree in Robotics and Autonomous Systems, with hands-on experience in robotics, automation, engineering design, and research. Experienced in designing robotic systems, developing prototypes, troubleshooting hardware and software, and taking projects from concept through implementation. Seeking a Summer 2027 engineering internship to contribute to challenging projects, gain industry experience, and grow as an engineer.

EDUCATION

Arizona State University - Polytechnic Campus, Mesa, AZ

B.S., Robotics and Autonomous Systems | Expected May 2028 | GPA: 3.8/4.0

TECHNICAL SKILLS

CAD/Design: Fusion 360, SolidWorks, Onshape, 3D printing software

Programming & Systems: Python, MATLAB, Embedded Systems (Raspberry Pi, ESP32, Arduino), Linux (Ubuntu & Arch), ROS2, C++, MuJoCo, Isaac Sim

Professional: Technical documentation, hardware integration, client/stakeholder communication, cross-functional collaboration

PROJECT & LEADERSHIP EXPERIENCE

MIT Auto-ID Lab — Research Assistant | *Jul 2026 – Present*

- Design and prototype low-cost, projectile-deployed environmental sensors for measuring temperature, moisture, and canopy microclimate conditions in remote forests.
- Contribute to the development of a drone-based deployment platform and low-impact attachment mechanism for sensor placement in hard-to-reach terrain.

LANL x ASU Devils Invent Hackathon — 1st Place, ASU | *Mar 2026*

- Designed and built a functional prototype within 48 hours, presenting the final system and technical results to Los Alamos National Laboratory judges.
- Engineered an automated system to safely scrape and collect fired Americium Dioxide from a crucible, reducing operator exposure and eliminating the need for skilled manual handling.

AMASS Lab — Research Assistant, ASU | *Jan 2025 – Dec 2025*

- Designed and built an autonomous blimp airframe for the “Defend the Republic” national competition, integrating structural, propulsion, and control systems.
- Programmed a Raspberry Pi in Python to control propulsion, applying embedded systems and control concepts to a functional autonomous prototype.

Sun Devil Motorsports — Team Member, ASU | *Sep 2024 – Present*

- Designed a cooling system for a Formula SAE vehicle and manufactured the vehicle's body harness using Rapid Harness.

ARMS Club — Vice President, ASU | *Aug 2026 – Present*

Advanced Robotics and Manufacturing Systems

- Help lead a student-run engineering consultancy delivering solutions for corporate clients.
- Oversee project execution and reinvest contract revenue into member-led R&D projects
- Created Digital Twins of Unitree Go2 and G1 using MuJoCo and Isaac Sim

WORK EXPERIENCE

Undergraduate Learning Assistant, RAS 101 — ASU | *Aug 2026 – Present*

- Guide students through Dobot programming, manipulation, and introductory robotics labs.
- Troubleshoot hardware and software issues while reinforcing robotics and mechatronics fundamentals.